

AMAN KUMAR SINGH

SENIOR SOFTWARE ENGINEER - AI

Bangalore, India · kumarsinghaman26@gmail.com · +91 81479 97054

linkedin.com/in/amanvimal · kumarsinghaman.github.io/Portfolio

PROFILE

Senior Software Engineer - AI with 7+ years building production-grade AI/ML and big-data systems for cybersecurity — Threat Detection (XDR), Bot Mitigation, and Fraud Analytics. Skilled in architecting and operationalizing low-latency, **cloud-native backend services** and ML pipelines on Azure and Databricks using **Python, Go, and Scala**, with hands-on Spark Streaming, ML/NLP, and generative AI. Passionate about leveraging generative, predictive, and agentic AI, distributed data processing, and cloud-native architectures to drive secure, scalable, high-impact AI products.

PROFESSIONAL EXPERIENCE

Senior Software Engineer - AI · Barracuda Networks — Bengaluru

Jun 2023 – Present

- Designing and implementing a **Zero-Day Attack Detection** framework using generative AI and adversarial ML, mapped to MITRE ATT&CK & ATLAS, to proactively surface unknown malware and exploit behaviors.
- Productionized a real-time **Data Exfiltration detection ML pipeline** for firewalls using behavioral baselines (connection / device / organization) with automated periodic retraining for data drift — improving XDR coverage and achieving **~80% true-positive rate** per customer feedback.
- Partnered with cross-functional teams to optimize the end-to-end Detection Engine for performance and cost — **cutting compute spend by ~60%** and eliminating periodic job restarts caused by out-of-memory failures.
- Led development of an AI-powered **Email Data Labeling Assistant** that automated labeling workflows and increased labeler productivity **~5x**, improving training-data quality for email-security models.
- Built a **Generative-AI WAF Smart Assistant** (OpenAI GPT-3.5 + RAG over product docs) to automate configuration, optimize settings from traffic patterns, generate real-time summaries and custom reports, and provide interactive chat support.

Tech: XDR, Cursor, Databricks, Delta Lake, MLflow (Feature Store & Model Registry), Databricks Asset Bundles (DABs), Tines, Zendesk, Retool, PostgreSQL, OpenAI GPT-3.5 (Text2SQL & doc search), CLIP, ChromaDB, NeMo Guardrails, Streamlit.

Software Engineer - AI · Barracuda Networks — Bengaluru

Jun 2021 – Jun 2023

- Built a cost-effective, cloud-native **Spark Structured Streaming pipeline** (stateless + stateful) on Azure from scratch for Advanced Bot Protection (WAF), processing **~1 TB/day** with multiple inline ML model predictions. Achieved an industry-low infrastructure cost of **~\$4 per million events** processed.
- Built **cloud-native monitoring microservices in Go** for the data pipeline and ML models, exposing **REST APIs** with alerting and auto-recovery to ensure high availability across Azure deployments.
- Productionized a **Client Similarity model** using Google's ScaNN at the application layer in WAF, enabling cross-customer threat correlation.

Tech: Scala, Go, PySpark, Spark MLlib, Azure Event Hubs, Databricks, Redis, Elasticsearch, Docker, Microservices (REST APIs), Flask, Spark Structured Streaming (stateless & stateful), ClickHouse.

Associate Software Engineer - AI · Barracuda Networks — Bengaluru

Jul 2019 – Jun 2021

- Developed and deployed a **Multi-Model ML-driven real-time Bot Detection** system using User Behavioral Analysis (UBA), blocking **~90% of malicious tool requests** across WAF customer sites.
- Built internal analytical dashboards using Redash to power data analysis for security and product teams.

Machine Learning Intern · HirePro — Bengaluru

Jan 2019 – Jun 2019

- Built a Candidate Voice Classifier (CNN) to detect voice impersonation in remote interviews.
- Built a Resume Classifier (multiple ML classifiers), Resume Parser (RegEx + NLP), and a Resume Splitter using a custom spaCy NER model to segment resumes by section.

AREAS OF EXPERTISE

Languages	Python, Go (Golang), Scala, SQL
AI / ML	Supervised/Unsupervised Models, Generative AI, LLMs, RAG, NLP, Agentic AI, MLflow, Spark MLlib, Google ScaNN, Model Retraining Pipelines (MLOps)
Data / Streaming	Apache Spark (Structured Streaming — stateless & stateful), PySpark, Kafka, Azure Event Hubs, Delta Lake, Elasticsearch, Redis, ClickHouse, PostgreSQL
Cloud / MLOps	Azure, Databricks, Databricks Asset Bundles (DABs), Docker, GitHub Actions
Tools / Frameworks	Tines, Retool, Redash, Streamlit, Flask, Jira, Zendesk, Cursor (AI-native coding & debugging)
Foundations	Data Structures & Algorithms, Distributed Systems, Cloud-Native Architecture, XDR (Extended Detection and Response)

EDUCATION

B.Tech, Computer Science — Reva University, Bengaluru · 2015 – 2019 · CGPA: 8.9 / 10

HONORS & AWARDS

- **Hackathon Winner**, Barracuda Networks (2024) — AI Email Data Labeling Assistant for faster, higher-quality ML training data.
- **Hackathon Winner**, Barracuda Networks (2023) — WAF Smart Assistant built on custom-trained OpenAI LLMs.
- **Cudathon Winner**, Barracuda Networks (2020) — Client Similarity model using Google ScaNN (Scalable Nearest Neighbors).
- **Hackathon Winner**, REVA University (2019); **HACKFEST Winner**, REVA University (2017).

LANGUAGES

English, Hindi